

agentic-ai-azure-sk-postgresql

Companion - Semantic Kernel + Azure PostgreSQL

Agentic AI on Azure - Enterprise Master Class | Handout (slides + notes)

Notes

Welcome. This deck covers 'agentic-ai-azure-sk-postgresql' - Companion - Semantic Kernel + Azure PostgreSQL. Frame the problem and why it matters, then walk the class through the learning goal, the enterprise use case, the architecture/flow, the key concepts, a live run in VS Code, and the architect's takeaways. The repo runs locally with no Azure, so demo it live.

Slide 2 - Learning goal

- Back an SK agent skill with a relational database
- Expose a native `@kernel_function` that reads orders
- Run offline; swap to Azure PostgreSQL via one env var

Notes

[Learning goal] Walk through these talking points:

- Back an SK agent skill with a relational database
- Expose a native `@kernel_function` that reads orders
- Run offline; swap to Azure PostgreSQL via one env var

Ask the class: which of these ideas is new to you?

Slide 3 - Enterprise use case

- Orders Assistant
- Answers 'where is my order ORD-1001?'
- Plugin reads an orders table (status, customer, total)

Notes

[Enterprise use case] Walk through these talking points:

- Orders Assistant
- Answers 'where is my order ORD-1001?'
- Plugin reads an orders table (status, customer, total)

Tip: relate this to a regulated scenario your students know.

Slide 4 - Architecture / flow

- Extract the order id from the question
- `kernel.invoke OrdersPlugin.get_order_status`
- Mock: in-memory table | Postgres: `SELECT` via `psycopg`
- Return a natural-language status

Notes

[Architecture / flow] Walk through these talking points:

- Extract the order id from the question
- `kernel.invoke OrdersPlugin.get_order_status`
- Mock: in-memory table | Postgres: `SELECT` via `psycopg`
- Return a natural-language status

Demo: trace a single request through these steps live.

Slide 5 - Key concepts

- @kernel_function + KernelArguments
- Data accessor branches mock vs psycpg
- POSTGRES_CONNECTION_STRING toggles the backend

Notes

[Key concepts] Walk through these talking points:

- @kernel_function + KernelArguments
 - Data accessor branches mock vs psycpg
 - POSTGRES_CONNECTION_STRING toggles the backend
- Open the named file in the repo while you explain each point.

Slide 6 - Run it (VS Code - no Azure needed)

- git clone the repo, then open it in VS Code
- python -m venv .venv -> activate it
- pip install -r requirements.txt
- uvicorn app.main:app --reload
- Open /docs and call POST /api/v1/orders/ask
- Runs in MOCK mode - no Azure/keys needed

Notes

[Run it (VS Code - no Azure needed)] Walk through these talking points:

- git clone the repo, then open it in VS Code
 - python -m venv .venv -> activate it
 - pip install -r requirements.txt
 - uvicorn app.main:app --reload
 - Open /docs and call POST /api/v1/orders/ask
 - Runs in MOCK mode - no Azure/keys needed
- Pause for questions before moving on.

Slide 7 - Architect's takeaways

- SK plugins are a clean seam to your data
- Keep SQL in the plugin, not the agent
- Use Managed Identity / sslmode=require in prod

Notes

[Architect's takeaways] Walk through these talking points:

- SK plugins are a clean seam to your data
- Keep SQL in the plugin, not the agent
- Use Managed Identity / sslmode=require in prod

Discussion: when would you NOT choose this approach?